SectraTEK

AI FOR CYBERSECURITY AND DIGITAL FORENSICS · LAB 07 COMPANION

Talk to the Evidence



Turn the forensic agent into a chatbot: ask a device image or an Android bug report in plain English, and get answers that cite their source.

Daniel Kwaku Ntiamoah Addai · Mobile Threat Research Lead, SectraTEK Lab · AI for Cybersecurity and Digital Forensics

THE IDEA

From a lab you run, to a chatbot you talk to

In the lab you built the agent step by step. Now we wrap that same agent in a chat window, so you (or a student, or a colleague) can just ask questions and watch it work.

IN THE LAB

You call run_agent(...)

Run cells, read the trace, build the tools by hand. Great for learning how it works.



AS A CHATBOT

You just type a question

A chat window drives the same agent. Good for demos, quick checks, and letting others try it.

TWO CHATBOTS

Two things you can point it at

device image

Ask the Device

Walks a full Android extraction and answers with the exact file it read.

"Which accounts are on this phone?" → reads the account database, cites accounts.db.

adb bugreport

Ask the Bug Report

Reads an Android bug report, which you can pull from your own phone, and cites the section.

"Which apps are installed?" → reads the package section, cites it.

WHY A BUG REPORT

Light enough to use your own phone

A full extraction needs special tools. An Android bug report is just a text file, so anyone can make one.

Lightweight

One sectioned text file: device info, apps, battery, Wi-Fi, accounts, logs.

From your own phone

Enable Developer Options, then run adb bugreport. No forensic kit needed.

Easy to search

The agent lists sections, reads one, or searches the whole report.

```
adb bugreport  mybugreport      # produces mybugreport.zip, then upload it to the chatbot
```

HOW IT WORKS

Same agent, with a chat window on top

The tools do the work

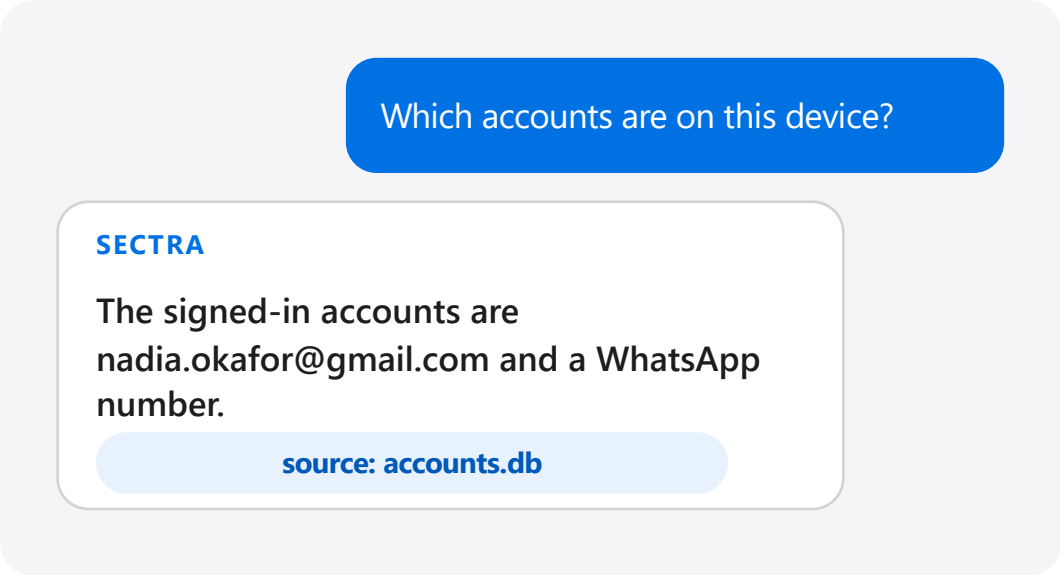
The agent lists, reads, and searches the evidence with read-only tools.

The loop runs the same

Think, act, observe, repeat, exactly as in the lab.

Every answer is cited

It names the file or section it used, so you can open and check it.



TWO WAYS TO RUN EACH

A web window, or right in the notebook

UI version

```
chatbot_ui.ipynb
```

Launches a real chat window (built with Gradio). In Colab you get a link you can open and share.

Best for demos and for people who just want to ask.

Jupyter version

```
chatbot_jupyter.ipynb
```

Call ask("...") in a cell and read the reasoning and the cited answer, no web UI.

Best for working through questions one at a time.

RUN IT ANYWHERE

Hosted in the cloud, or private on your PC

hosted Qwen

In Colab

Open the notebook, paste a free OpenRouter key, and go. Nothing to install.

- No setup
- Free model
- Shareable link

local Qwen (Ollama)

On your PC

Run a Qwen model locally with Ollama. No API key, and nothing leaves your machine.

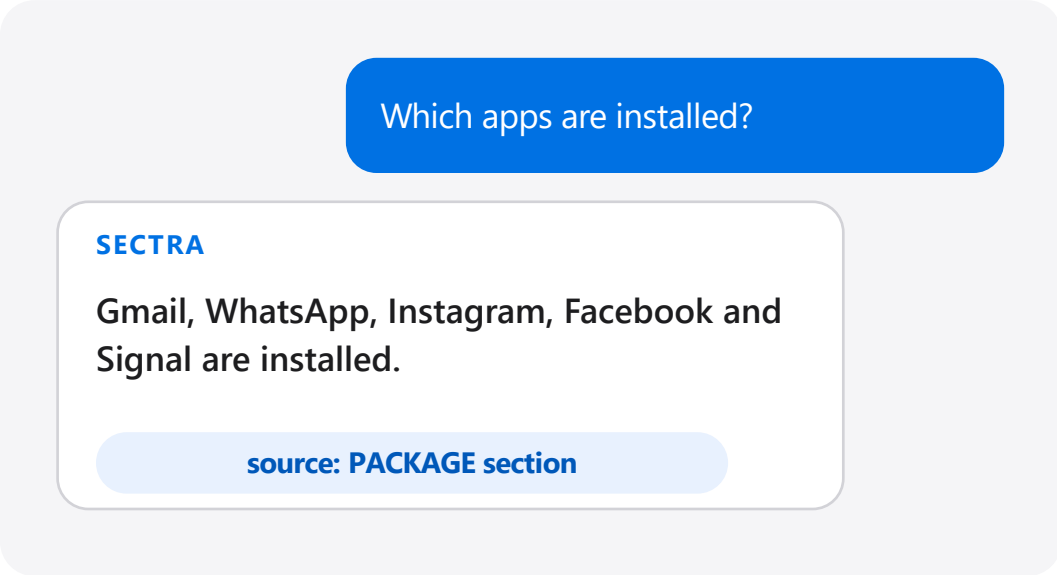
- No key
- Fully offline
- Private

USE IT

Ask in plain English, get a cited answer

THINGS TO ASK

- Which email accounts are signed in on this device?
- Which apps are installed?
- What is the most recent text that was sent?
- What Wi-Fi networks are saved?
- Any errors or crashes in the log?



Every answer names the file or section behind it, so you can verify it yourself.

YOUR TURN

Open a chatbot and ask it something

1	**Open a chatbot notebook**	chatbot_ui.ipynb (device) or bugreport_chatbot_ui.ipynb (bug report).
2	**Paste a free key**	A free OpenRouter key, or switch to local Ollama for no key.
3	**Launch the chat**	Run the last cell. In Colab, open the link it prints.
4	**Load your evidence**	Use the sample, or upload your own adb bugreport.
5	**Ask away**	Type a question, read the answer, and check the cited source.

SectraTEK

Ask, and check the source.

The chatbot makes the agent conversational, and still shows the exact file or section behind every answer.

